JALOLIDDIN KULMATOV (JAKE)

AI / ML Engineer · Full-Stack Developer · Founder of ICOM

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PROFILE

Engineering student at Jeonbuk National University with a 4.47/4.5 GPA and a full scholarship. Passionate about science and technology, with a strong interest in web development, machine learning (ML), data science and problem-solving. Founder of ICOM, an AI platform for international students in Korea (patent, conference paper).

EDUCATION

Jeonbuk National University (JBNU) *B.S. International Engineering and Science* — Jeonju, South Korea

- GPA 4.47/4.5. Full scholarship. Ranked 1st of 69 international students in Engineering.
- **JBNU President's Award**: the only engineering student with a perfect 4.5/4.5 GPA for a full year.

Universiti Malaya (UM) *Exchange student* — Sep 2026 – Feb 2027

- Courses: Probability & Statistics, Soft Computing, Control Systems, Software Modelling, Robotics, German I.
- Earlier: one semester at New Uzbekistan University, with a full scholarship.

EXPERIENCE

Founder & Full-Stack Developer · *ICOM (icom.ai.kr)* — Apr 2026 – Present

- Built the whole platform by myself. It helps international students in Korea with chat, internships, visas, housing, banking and clubs, and has an AI assistant.
- Front end in Next.js and TypeScript. Back end in Flask with a PostgreSQL database. Hosted on Vercel and Render.
- Built a system that collects internship posts from Wanted.co.kr twice a day and translates them with AI.
- Reached 50+ users at JBNU. Got a patent. Wrote a conference paper with Prof. Aaron Snowberger.

Undergraduate Research Assistant · *Solar Cell Lab, JBNU* — 2026 (4 months)

- Helped PhD students simulate and improve solar cells in COMSOL and PC1D. Helped prepare research papers.
- Built SCAPS IV Extractor, a web tool that turns SCAPS-1D simulation files into one Excel table (Voc, Jsc, FF, PCE).

Python & Full-Stack Developer · *Freelance and personal projects* — Mar 2025 – Present

- Built 50+ apps and websites, including an object detection system and a strawberry ripeness classifier. 80 public projects on GitHub.
- Connected apps to REST APIs. Analyzed and charted data with Pandas, NumPy, Matplotlib and Plotly.

IELTS Teacher & Content Manager · *5 Skills Educational Center, Uzbekistan* — May 2024 – Feb 2025

- Made lessons for each student. Ran full mock exams, including speaking tests.
- Managed the center's website. Helped translate and edit IELTS books, including *IELTS Listening Practice* by Hoshinori Matsuzono.

SELECTED PROJECTS

ICOM · International Community in Korea · icom.ai.kr · GitHub — 2026

One platform for international students in Korea: chat, internships, visa guides, housing, banking and an AI assistant. New internship posts are added and translated twice a day. Next.js 14, TypeScript, Tailwind, Flask, SQLAlchemy, JWT, PostgreSQL (Neon), APScheduler, Vercel, Render

TALABA · platform for students in Uzbekistan · talaba-zeta.vercel.app — 2026

ICOM for students in Uzbekistan, in Uzbek. It covers community, jobs, university admission, housing, scholarships and an AI assistant. Starting at the National University of Uzbekistan. Next.js, TypeScript, Tailwind, Python back end

Storyling · language learning through graded stories — 2026, in progress

A reading app for 10 languages. It picks stories that match the words you already know. Tap a word to see its meaning, listen to audio, and review new words. Planned the product for web, iPhone and Android. TypeScript monorepo: Next.js, Expo (React Native), shared learning engine, FSRS

Nori (노리다) · AI résumé converter for jobs in Korea 2026
Upload an English CV and get a Korean-style résumé PDF. AI translates and reorganizes it. Flask, Groq (Llama 3.3 70B), pdfplumber, ReportLab

VoiceMimic · voice cloning web app 2026
Record or upload a short voice clip, type any text, and hear it in that voice. Python, Flask, Coqui XTTS-v2, JavaScript audio recording

C-Test Trainer · German onSET exam practice · ctest-trainer-eight.vercel.app 2026
Practice site for the German onSET test. It has 4 full timed tests, 24 grammar topics, 593 vocabulary words, a mistake notebook and progress charts. HTML, CSS, JavaScript

More on GitHub: Find_me (finds a username on 40+ sites) · Food Reels (hackathon app: watch food videos and order) · Flask blog (login, comments) · house price regression model.

PROGRAMS & LEADERSHIP

Ambassador & Fellow · *Mongolian Society of AI in Medicine* Jan 2026 – Present

- Work with members from 7 countries on AI for healthcare in Central Asia, Mongolia and Korea. Finished the 6-month Silk Road Health Data Science Fellowship (Feb–Aug 2026).

Vlogger · *Hello_JBNU Broadcasting & Media Team* 2026 – Present

- Make videos about university and student life.

SAT Math Support Teacher *(volunteer)* May 2023 – Sep 2023

- Helped 30 students reach SAT Math scores of 700–760. Guided them through university applications.

PUBLICATIONS & AWARDS

- **Paper:** Kulmatov, J., & Snowberger, A. "Design and Implementation of an AI-Assisted Information and Community Platform for International Students in Korea." 2026 Summer Conference, Korean Society of Quality of Life Informatics / Korean Society of Physical AI.
- **Patent** from the ICOM project (2026).
- **Winner**, UNOSD Summer Camp, Seoul (2026): team study of the Bonyic Hydroelectric CDM Project, with students from 16 countries.
- **Finalist**, regional AI Hackathon, South Korea.
- **National selection stage**, International Junior Science Olympiad (IJSO), Uzbekistan. Regional Chemistry Olympiad winner (several years).

SKILLS

Programming	Python, TypeScript, JavaScript, C, C++, SQL, HTML/CSS
AI / ML	LLM APIs (OpenAI, Groq/Llama), function calling, embeddings, Hugging Face, voice cloning (XTTS), computer vision, regression modelling
Data	NumPy, Pandas, Matplotlib, Seaborn, Plotly, Jupyter
Web & cloud	Next.js, React, Tailwind, Flask, FastAPI, SQLAlchemy, PostgreSQL, REST APIs, JWT, Git/GitHub, Vercel, Render, Neon
Engineering	COMSOL, PC1D, SCAPS-1D
Languages	Uzbek (native), English (IELTS 7.5), Korean (TOPIK 4), Russian (intermediate), German (beginner)